Tetrad-Based Departure Decomposition for FLRW Anisotropy

Final Results Report — Credible-Claim Distillation

BASS / HTT / MIO / OBSSTAT stack

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Compiled June 26, 2026

Abstract

This report distills the *credible, defensible* results of the tetrad-based departure-decomposition program into a single self-contained document. It deliberately excludes development history, superseded paths, and any result that does not survive the project's claim-firewall. Every result below is one of four kinds: (i) framework statements about the master departure comparator $x_C = \Sigma^2_{\rm std} - W^2_{\rm std} + \Omega_{\rm tilt} + \Omega_{k,{\rm aniso}}$ and its claim-tiered diagnostic variables; (ii) conditional analytic theorems—both the EGS-type low-ℓ identities and a multicomponent identifiability and restricted Bianchi-I dynamics suite—each proven symbolically with the Wolfram Engine under explicitly stated hypotheses; (iii) model-independent measurements on real Planck and Cosmicflows-4 data, each calibrated against an isotropic null, a forward mock, or a resampling baseline; and (iv) explicitly labelled transfer-conditional diagnostics. The analytic suites are hardened by an external adversarial audit: every theorem is restated under its exact hypotheses (closure-conditional identities, an estimator sampling-variance statement rather than a universal floor, and an additive depth-transport contrast), the restricted Bianchi-I dynamics are re-derived by an *independent* chain-rule verifier (1000-state conservation and Gauss/Codazzi transport, DOP853/Radau cross-checks), and the estimator mechanics (hierarchical bulk-flow coverage, a curl-suppression non-identifiability no-go, and a look-elsewhere max-scan calibration) are validated on synthetic catalogues with known truth. No result here identifies a Bianchi family, detects an anisotropic geometry, claims native low-ℓ Boltzmann-solver validation, or promotes any diagnostic into a posterior-odds or model-ranking statement. The honest publishable envelope is a methods-plus-bounds framework that recovers the established low-ℓ CMB anomalies as null-calibrated model-independent features, maps the CF4 bulk-flow apex profile, measures a forward-mock-calibrated CF4 bulk flow and its affine velocity-gradient decomposition, and proves an exact dust-FLRW constraint-transport oracle (machine-checked with the Wolfram Engine and an independent xAct canonicalization)—while producing no false anisotropy claim.

Contents

1 Scope and claim discipline

This document is a *final-results distillation*, not a record of how the results were obtained. The full multi-chapter VER06 manuscript (which retains derivations, audit trail, and conditioned legacy appendices) is preserved separately and unchanged. Here we carry only what is currently defensible.

Claim tiers. All current public results sit in a claim-tiered observatory. We use the project's tiers: **C1** framework / disclosure, **C2** transfer-conditional, **C3** observer-side diagnostic, **C4** local/global discrimination candidate (gated). Every result in this report is *diagnostic-only* or *transfer-conditional*; none is a production-validated or geometry-level claim.

What this report does *not* claim. The following remain blocked and are stated here once, up front, so that no later sentence can be read as lifting them:

- no identified Bianchi family and no detected anisotropic geometry;
- no native low-ℓ Bianchi Boltzmann-solver output and no native morphology atlas (both absent in this repository state);
- no promotion of any scalar diagnostic (x_C, Q, Π, F, G_F), direction coherence, or low-ℓ feature into Bianchi-family or geometry evidence;
- no posterior odds, Bayes-factor headline, or model ranking; MIO observatory certificates are diagnostic-only and are kept separate from HTT-owned inference;
- no native-validation label on any external/proxy transfer output.

These blocks are lifted only by future work supplying a native low-ℓ morphology atlas with matched masks, matched randoms, full covariance, calibrated nulls, predictive checks, and family-equivalence gates. None of that is present here, and nothing below depends on it.

2 The tetrad departure framework

Master departure comparator. The program organises FLRW departure through a single signed comparator coordinate,

$$x_C \;=\; \Sigma^2_{\rm std} \;-\; W^2_{\rm std} \;+\; \Omega_{\rm tilt} \;+\; \Omega_{k,{\rm aniso}}, \tag{1}$$

where $\Sigma^2_{\rm std}$ is a standardised shear-squared contribution, $W^2_{\rm std}$ a standardised vorticity-squared contribution, $\Omega_{\rm tilt}$ a tilt (peculiar-rapidity) term, and $\Omega_{k,{\rm aniso}}$ an anisotropic spatial curvature term, each referred to a stated frame (normal, matter, CMB, local-boost, or geometry frame).

Definition 1 (Comparator, not magnitude)**.** x_C in (1) is a *signed* comparator coordinate built to compare a model against the FLRW point, not an invariant anisotropy magnitude. Its sign structure (note the $-W^2_{\rm std}$) means x_C may cancel or change sign; it must never be read as a nonnegative "total anisotropy."

Claim-tiered diagnostic variables. On top of (1) the observatory exposes four observer-side scalars, all diagnostic-only:

- *occupancy* Q, a policy-normalised departure score;
- *exceedance* Π, an empirical exceedance curve of a named score above stated thresholds;
- *filling* $F = x_C/x_{\max}$, the fraction of an algebraic admissible ceiling $x_{\max}$ occupied by the comparator;
- *depth gap* $G_F(z) = F(z)/F(z_{\mathrm{ref}})$, the redshift-resolved ratio of the filling.

The algebraic ceiling used throughout is the S2a value $x_{\max} = 9.25 \times 10^{-6}$, the combined shear+tilt+curvature admissible ceiling for the full comparator x_C; it is distinct from, and larger than, the VER06 manuscript's shear-only bound $\Sigma^2_{\mathrm{std}} < 6.4 \times 10^{-6}$ (the latter bounds Σ^2_{std} alone, the former the whole signed comparator). These scalars summarise a comparator; on their own they do not classify a geometry or a Bianchi family.

Kinematic-bound linkage. The comparator is tied to the Maartens–Ellis–Stoeger (MES) kinematic-bound hierarchy: bounds on CMB multipole anisotropy bound the covariant shear, vorticity, and acceleration, which in turn bound the contributions in (1). We use MES as a derived bound under its stated hypotheses (almost-isotropy of the CMB in a family of observers), not as an asserted constraint.

3 Conditional analytic theorems (Wolfram-verified)

The Ehlers–Geren–Sachs (EGS) theorem states that if the CMB is exactly isotropic about a congruence of geodesic observers in an expanding dust universe, the spacetime is FLRW; the Stoeger–Maartens–Ellis "almost-EGS" result makes this stable. The three theorems below carry that chain onto the diagnostic variables of Section 2. Each is a structural or limiting statement under explicitly stated hypotheses—*not* a detection—and each algebraic step, limit, and variance propagation was verified symbolically with the Wolfram Engine (14.3.0). The machine-checked proof record is `docs/generated/egs_lowell_theorem_proofs.json`. We write the temperature-quadrupole magnitude a_2, the shear scalar Σ, the quadrupole power $D_2 \propto a_2^2$, the algebraic ceiling $x_{\max}$, and the shear filling $F_{\mathrm{shear}} = \Sigma^2/x_{\max}$.

Theorem 1 (Quadrupole–filling closure-conditional identity (NT-A1))**.** *Assume the free-streaming linear $\ell = 2$ closure $a_2 = \kappa\,\Sigma$ with $\kappa > 0$ a constant symbolic coefficient (the Ehlers–Treciokas–Matravers value $\kappa = 4/21$ is used only for the figure and is not load-bearing; the identity holds for any $\kappa > 0$), $F_{\mathrm{shear}} = \Sigma^2/x_{\max}$, and $D_2 = c_D\, a_2^2$ with $c_D > 0$, and that the tilt rapidity β enters the dipole a_1 only. Then*

$$F_{\mathrm{shear}} \;=\; \frac{a_2^2}{\kappa^2\, x_{\max}} \;=\; \frac{D_2}{c_D\, \kappa^2\, x_{\max}} \;\propto\; D_2, \tag{2}$$

so $F_{\mathrm{shear}} \to 0$ as $D_2 \to 0$ (the isotropic-quadrupole limit: an isotropic CMB quadrupole forces zero shear-filling under this closure), and $\partial F_{\mathrm{shear}}/\partial\beta = 0$ (the tilt sector is invisible to the shear-filling). This is a closure-conditional identity, not an EGS theorem.

Proof. Substitute $\Sigma = a_2/\kappa$ into $F_{\mathrm{shear}} = \Sigma^2/x_{\max}$ and $a_2 = \sqrt{D_2/c_D}$; the limit and the β-derivative are immediate. The $\ell = 2$ projection that licenses the closure rests on the sphere averages $\langle n_a n_b\rangle = \frac{1}{3}\delta_{ab}$ (Wolfram-verified). Conditional on the free-streaming linear closure. □

Theorem 2 (Single-sky sampling variance of the shear-filling estimator (NT-A3))**.** *Under Theorem 1, F_{shear} is linear in the quadrupole power C_2. For the standard ideal full-sky Gaussian power* *estimator $\hat{C}_2$ with single-sky sampling variance* $\mathrm{Var}(\hat{C}_2) = 2\,C_2^2/(2\ell+1)$ *(χ^2 with $2\ell+1$ degrees of freedom),*

$$\frac{\sigma(\widehat{F_{\rm shear}})}{F_{\rm shear}} = \sqrt{\frac{2}{2\ell+1}} \xrightarrow{\ell=2} \sqrt{\tfrac{2}{5}} \approx 0.632. \tag{3}$$

Hence this estimator has $\sim 63\%$ fractional sampling dispersion at the quadrupole. This is the sampling variance of one specific (ideal full-sky Gaussian) estimator, not *an estimator-optimal lower bound or a universal floor over all estimators; an information-bound version would require a stated likelihood, parameter space, estimator class, Fisher information and regularity conditions.*

Proof. Error propagation for the stated estimator, $\mathrm{Var}(\widehat{F_{\rm shear}}) = (\partial F_{\rm shear}/\partial C_2)^2\,\mathrm{Var}(\hat{C}_2) = \frac{2}{2\ell+1}\,F_{\rm shear}^2$; take the square root and set $\ell = 2$ (Wolfram-verified, branch-free squared identity). □

Theorem 3 (Depth-transport identity for the filling (NT-B3))**.** *Write the depth-resolved filling additively, $F(z) = F_{\rm shear}(z) + F_{\rm tilt}(z)$. The primary depth observable is the additive contrast $\Delta_F(z) = (LF)(z) \equiv F(z) - F(z_{\rm ref})$, where the difference operator L annihilates constants, $L\,\mathbf{1} = 0$. For a depth-steady shear with no tilt, $\Delta_F(z) \equiv 0$ (no depth contrast); a depth-evolving tilt imprints a nonzero $\mathrm{d}\Delta_F/\mathrm{d}z$. The legacy depth ratio $G_F(z) = F(z)/F(z_{\rm ref})$ is recovered only on the branch $F(z_{\rm ref}) \neq 0$ (it is $0/0$-indeterminate otherwise).*

Proof. With $F_{\rm tilt} \equiv 0$ and $F_{\rm shear}$ constant in z, $\Delta_F = F - F = 0$ and $\mathrm{d}\Delta_F/\mathrm{d}z = 0$; a power-law tilt $F_{\rm tilt} = c\,z^q$ gives $\mathrm{d}\Delta_F/\mathrm{d}z = c\,q\,z^{q-1} \neq 0$ (Wolfram-verified). Conditional on the additive decomposition. A nonzero Δ_F trend is depth evolution of the filling; attributing it specifically to the tilt sector requires a separate projected response-rank gate. □

Reading. The three theorems make precise, in the program's own variables, three EGS-type (closure-conditional) statements: an isotropic CMB quadrupole forces zero shear-filling under the linear closure and the tilt sector cannot counterfeit it (NT-A1); the standard single-sky quadrupole-power estimator carries $\sim 63\%$ fractional sampling dispersion (NT-A3, the sampling variance of that estimator, not a universal floor); and a depth-steady shear leaves no depth contrast, so any measured $\Delta_F \neq 0$ trend is depth evolution of the filling (NT-B3), attributable to the tilt sector only with a separate response-rank gate. All are conditional structural statements; $\kappa = 4/21$ is the cited ETM value used only for the figure and the theorems hold for any symbolic $\kappa > 0$.

Broader conditional-theorem registry. A wider set of program theorems (labelled B4, G2, G5, S1, S3, S4, S5) carries kinetic-theory and information-bound statements onto the same variables. These are recorded as *convention-conditional* or *program* theorems whose observational use is blocked by explicit kill-switches (e.g. a nonpositive collision gap blocks any exponential-forgetting language; a near-zero source rank blocks any inverse-source claim). They are synthetic/manufactured verification only and are not promoted to observational results here; the auditable registry is `docs/generated/theorem_extension_registry.json`.

4 Multicomponent identifiability and restricted Bianchi-I dynamics

A second conditional-theorem suite (the PAPER-A/PAPER-B program) sits on the multicomponent canonical bridge: the covariant congruence first jet, the canonical symmetric–trace-free (STF) multicomponent state, and a restricted Bianchi-I multifluid moment sector. Seven analytic cores were verified symbolically with the Wolfram Engine; the machine-checked record is `docs/generated/pr04_paper_theorem_proofs.json`, and the numerical companions are the 23 property gates under `research_gates/pr04/tests/` (run by `make paper-a-gates` / `make paper-b-gates`). Each statement is conditional on its registered hypotheses—it is a structural identity, not a detection.

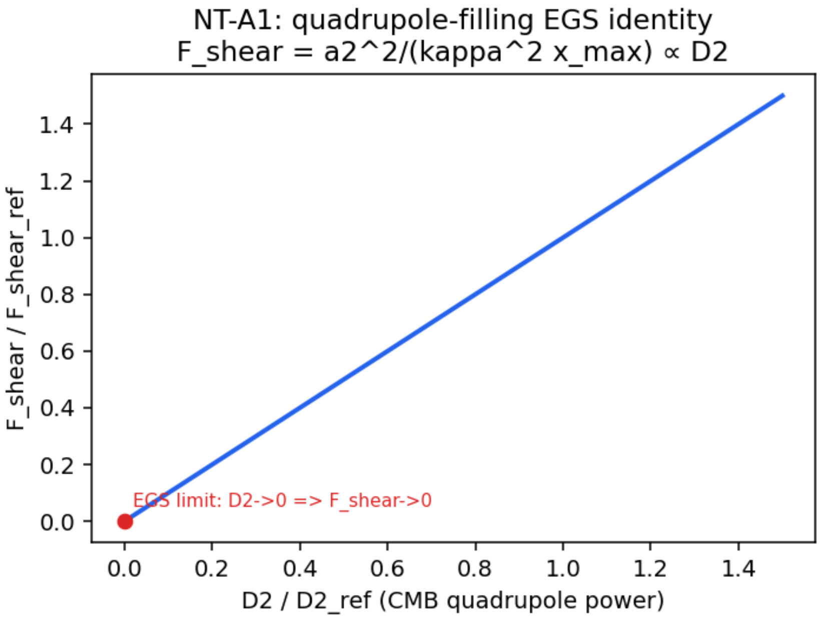

Figure 1: Theorem 1 (NT-A1): the shear filling fraction is linear in the CMB quadrupole power D_2 and vanishes in the isotropic-quadrupole limit $D_2 \to 0$ under the linear closure. Diagnostic-only analytic illustration; not a detection.

PAPER-A (identifiability and congruence kinematics). Write a whitened response matrix stacked from blocks, with M the velocity-gradient tensor of a congruence.

- *A-rank.* The identifiable dimension is the rank of the whitened stacked response; appending a duplicate block adds *zero* rank and places the duplicated directions on the $(x, -x)$ nullspace. The null space *equals* that $(x, -x)$ line only when the block has full column rank; otherwise it merely contains it. Identifiability is therefore a rank property of the response design, not of the data magnitude.
- *A-flrw.* A flat-FLRW comoving congruence has expansion $\theta = 3H$ and exactly zero shear and acceleration.
- *A-boost (composition).* Two non-collinear boosts compose to an exact Lorentz transformation whose velocity is *not* the Euclidean sum of the two boost velocities: rapidities do not add as vectors. This is an exact composition / non-Euclidean velocity-addition statement; it is *not* a Wigner-rotation theorem (no decomposition into a net boost and spatial rotation, and no rotation angle, is claimed here).
- *A-first-jet (no-go).* Equality of the pointwise four-velocity of two congruences does not determine its first derivative: two fields can share u^a at a point yet differ in expansion there. Hence a congruence first jet—not a single rapidity—is needed to define expansion, shear, and vorticity.
- *A-radial-novortex (radial-vorticity no-go).* For a purely radial response at directions n and depths d (so $r = d\,n$) and any antisymmetric affine mode Ω, the radial response $n^a \Omega_{ab} n^b = 0$ exactly: vorticity lies in the data null space. Radial peculiar-velocity data therefore carry no vorticity information.
- *A-shell-degeneracy.* A single-depth-shell dipole design (acceleration-like block n_i plus coherent-bulk block n_i/d) has rank 3; restoring broad depth support lifts it to the full rank 6. One shell cannot separate acceleration from bulk.
- *A-temporal-rank.* For an STF tensor history with temporal kernel T, the dynamic design has

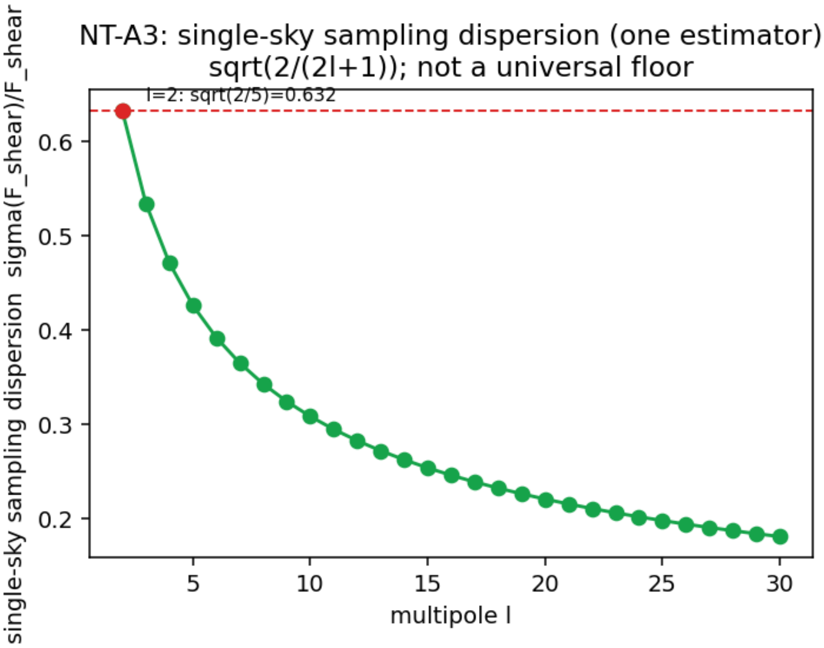

Figure 2: Theorem 2 (NT-A3): the single-sky fractional sampling dispersion $\sqrt{2/(2\ell+1)}$ of the standard full-sky F_{shear} estimator, equal to $\sqrt{2/5} \approx 0.632$ at the quadrupole (not a universal floor). Diagnostic-only; not a detection.

rank $5\,\text{rank}(T)$ (i.e. rank 15 for three independent temporal rows): tensor observability requires independent temporal kernels.

The last three were the previously open PAPER-A corollaries; they are now closed by independent matrix proofs (`htt/.../departure/paper_a_closure.py`) plus the `research_gates/pr07/` property gates and the Wolfram symbolic core.

PAPER-B (restricted Bianchi-I multifluid). Branch (stated before each theorem): expanding geodesic normal congruence, Fermi-propagated orthonormal triad, flat Bianchi-I, homogeneous non-interacting perfect-fluid species ($\hat{p} = w\hat{\rho}$, $-1 < w \le 1$), signature $(-,+,+,+)$, $c = 1$. Write the tilt second moment K, its trace $\Omega_{\text{tilt}} = \operatorname{tr} K$, its physical STF anisotropic stress π_{ab}, and the dimensionless normalized moment $\Pi_{ab} = \kappa\pi_{ab}/(3H^2)$.

- *B-nonsuff (scalar non-sufficiency).* A colinear antipodal pair and an isotropic six-stream can share the same scalar Ω_{tilt} while differing in the STF moment Π; the scalar trace alone cannot close the tilt sector.
- *B-psd (moment cone).* Every positive-semidefinite second moment $K = \sum_i \lambda_i e_i e_i^\top$ is realized by antipodal stream pairs, each with exactly zero first moment; the realizable moment cone is the PSD cone.
- *B-dust (exact FLRW oracle + constraint transport).* The closed-form dust scale factor $a(t) = (1 + \frac{3}{2}H_0 t)^{2/3}$ gives $H = H_0/(1 + \frac{3}{2}H_0 t)$ and the exact Raychaudhuri relation $\dot{H} = -\frac{3}{2}H^2$ with $3H^2 = \kappa\mu$ (Gauss) and zero shear. The RK4 integrator reproduces it to $\sim 2 \times 10^{-9}$ with a Gauss residual below 2×10^{-8}. This is the exact FLRW comparator for the constraint-transport gate.
- *B-shear (dimensionally consistent shear memory).* Shear evolves by $\dot{\sigma}_{ab} = \text{STF}(-3H\sigma_{ab} + \kappa\pi_{ab}) = \text{STF}(-3H\sigma_{ab} + 3H^2\Pi_{ab})$, with $\partial\dot{\sigma}/\partial\pi = \kappa$ and $\partial\dot{\sigma}/\partial\Pi = 3H^2$ (both nonzero), so ablating the anisotropic stress provably changes the shear history: shear retains memory of the

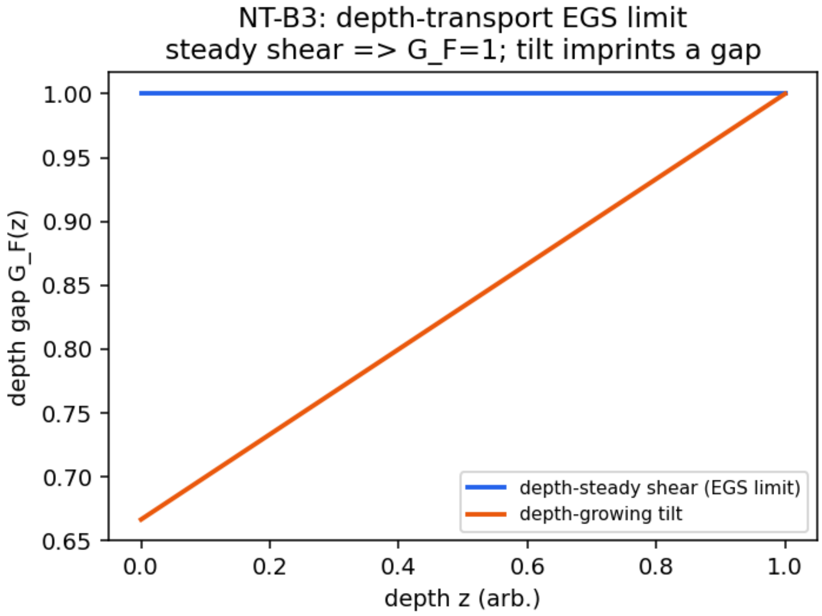

Figure 3: Theorem 3 (NT-B3): the depth contrast $\Delta_F(z) = F(z) - F(z_{\rm ref})$ is flat at zero for a depth-steady shear and departs from zero when a depth-evolving tilt is present. Diagnostic-only; not a detection.

tilt second moment. The two source forms are exactly equivalent; the dimensionally invalid form $\kappa\Pi$ is not used.

Independent dynamics verification. The restricted Bianchi-I dynamics are validated by a verifier that does *not* call the production right-hand side and compare it with itself. It samples admissible species (rest-frame density, equation of state, subluminal velocity), projects *independently* to the normal-frame (μ, p, q, π), differentiates the projection by the chain rule, and compares against the analytic energy/momentum conservation laws. Over 1000 seeded admissible states the maximum energy residual is 5.7×10^{-14} and momentum residual 5.1×10^{-14}; the Gauss and Codazzi constraint-transport residuals are 5.6×10^{-17} and 0. The exact $\Lambda = 0$ dust oracle is reproduced by SciPy DOP853 (max $|\Delta a| = 9.6 \times 10^{-13}$) and Radau ($3.2 \times 10^{-14}$), and the RK4 step shows fourth-order convergence. These cross-checks (the `research_gates/pr07/` dynamics gates) close the constraint-transport and conservation review as an *independent* re-derivation, not a single production fixture.

Symbolic and chain-of-verification gate surface. The closed-form analytic cores carry a machine-checked symbolic record. The Wolfram/xAct gate (`make pr07-wolfram`→`docs/generated/pr07_wolfr` verifies, all checks true (engine backend `wolframclient`, xAct 1.3.0): the physical/normalized shear-source equivalence with derivatives κ and $3H^2$; exact Lorentz composition with non-additive velocity; the first-jet and radial-vorticity identities; the dust Raychaudhuri relation; and an independent coordinate derivation of $\theta = H_1 + H_2 + H_3$ (cross-checked against the abstract-index xAct canonicalization). A concise chain-of-verification report (`python scripts/cove_verify_pr07.py`→`pr07_cove_report.json`) aggregates the theorem and synthetic mechanics into 13 evidence checks, all passing under the registered delegations (the data/E2E/solver blockers of §8).

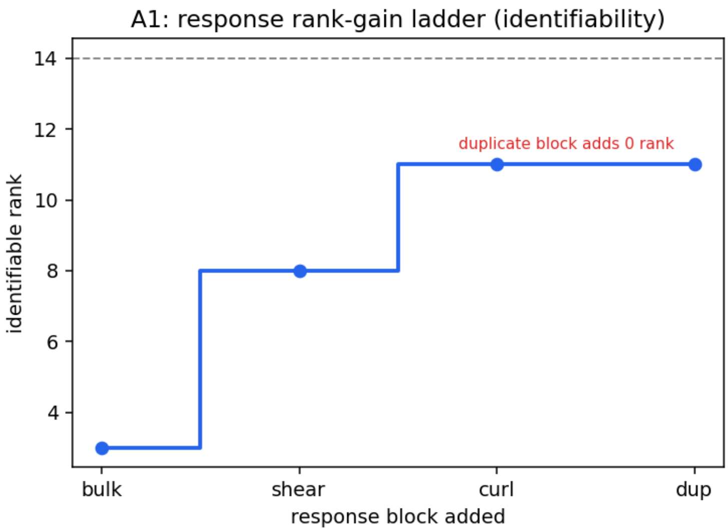

Figure 4: A-rank: the identifiable rank grows as complementary response blocks are added and is unchanged by a duplicate block (the duplicated directions occupy the nullspace). Diagnostic-only; not a detection.

5 Model-independent real-data measurements

Four model-independent measurements are produced from real Planck and Cosmicflows-4 data, using the observer-side estimator stack. All are diagnostic-only: each is calibrated against an isotropic null, a forward mock, or a resampling baseline, and none produces a Bianchi-family, geometry, native-solver, posterior, or certificate claim. Two (K1, K4) summarise the CMB low-ℓ map and the CF4 reconstruction-grid apex profile; two (K5, K6) use the full CF4 group release and the affine velocity-gradient decomposition.

5.1 K1 — Null-calibrated low-ℓ morphology of the real Planck map

We summarise the Planck PR3 SMICA temperature map at NSIDE$=16$ with a set of low-ℓ scalar and morphology statistics, and calibrate each against an isotropic ΛCDM `synfast` null ensemble ($n = 10000$, seed 12345, CAMB reference spectrum). Table 1 gives the observed values and the null tail probabilities.

Table 1: K1: null-calibrated low-ℓ statistics of the Planck PR3 SMICA NSIDE$=16$ map. p is the tail probability against the isotropic ΛCDM null ($n = 10000$). Look-elsewhere across the six statistics is tracked, not globally corrected.

Statistic	Observed	p	Tail
$S_{1/2}$	6.10×10^3	0.059	$P(\text{null} \leq \text{obs})$
parity even/odd ratio	0.628	0.018	$P(\text{null} \leq \text{obs})$
parity asymmetry	-0.228	0.018	$P(\text{null} \leq \text{obs})$
planarity mean	0.518	0.043	$P(\text{null} \geq \text{obs})$
$\ell2$–$\ell3$ axis alignment	69.2°	0.65	$P(\text{null} \leq \text{obs})$
axis-to-CMB-dipole apex	72.9°	0.71	$P(\text{null} \leq \text{obs})$

Findings. The map recovers the parity (odd-power excess, $p = 0.018$) and large-angle ($S_{1/2}$, $p = 0.059$) low-ℓ anomalies as model-independent, null-calibrated features; planarity is marginal

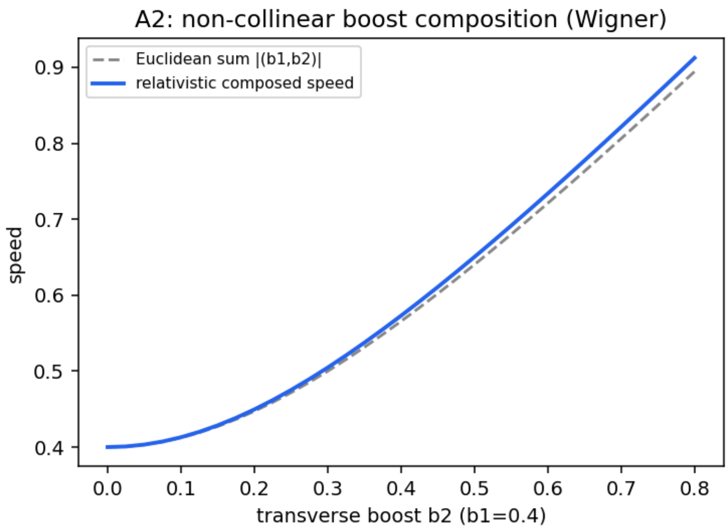

Figure 5: A-boost-composition: the relativistic composed speed of two non-collinear boosts departs from the Euclidean sum of the boost velocities (exact Lorentz composition, not a Wigner-angle claim). The separate A-first-jet no-go establishes that pointwise four-velocity does not determine congruence kinematics. Diagnostic-only analytic illustration.

($p = 0.043$). The power-inertia $\ell 2$–$\ell 3$ axis and the axis-to-CMB-apex angle show *no* anomaly in this estimator ($p \approx 0.65, 0.71$); these are honest nulls, and this is not the multipole-vector "axis-of-evil" statistic. The preferred low-ℓ axis lies at Galactic $(l, b) = (326.6°, -1.0°)$. Because the look-elsewhere effect across the six statistics is tracked but not globally corrected, the parity $p = 0.018$ over six trials is not globally significant. No Bianchi or source-identification claim is made. The map is full-sky cleaned with no mask deconvolution or full pixel–pixel covariance.

5.2 K4 — CF4++ bulk-flow apex versus depth

From the Cosmicflows-4 (CF4) reconstruction grid we measure the volume-averaged bulk-flow amplitude and apex direction in seven radial shells. Table 2 gives the profile.

Table 2: K4: CF4++ bulk-flow apex by depth shell. $|B|$ is the volume-averaged reconstruction velocity per shell; the bracketed figures are bootstrap apex dispersion over correlated reconstruction cells, which under-represents the direction uncertainty and is not a full covariance. Full-sample bulk flow: 133.4 km/s toward $(l, b) = (121.6°, 70.8°)$.

r [Mpc/h]	N cells	$\|B\|$ [km/s]	apex (l, b) [deg]	to CMB apex [deg]
90–150	6749	378	$(300, 24)$	36.9
150–200	10254	531	$(301, 63)$	25.3
200–250	16091	584	$(239, 88)$	39.9
250–300	21788	588	$(133, 75)$	52.5
300–350	27890	222	$(114, 48)$	80.4
350–400	35806	128	$(114, -49)$	160.6
400–450	45179	186	$(120, -78)$	147.5

Findings. The bulk-flow amplitude rises to ~ 590 km/s at 250–300 Mpc/h and then falls to ~ 130–190 km/s in the deep shells. The apex stays within ~ 5–46° of the full-sample apex out to 300 Mpc/h and then reverses (~ 120–149°) in the deepest, sparsest shells, where the

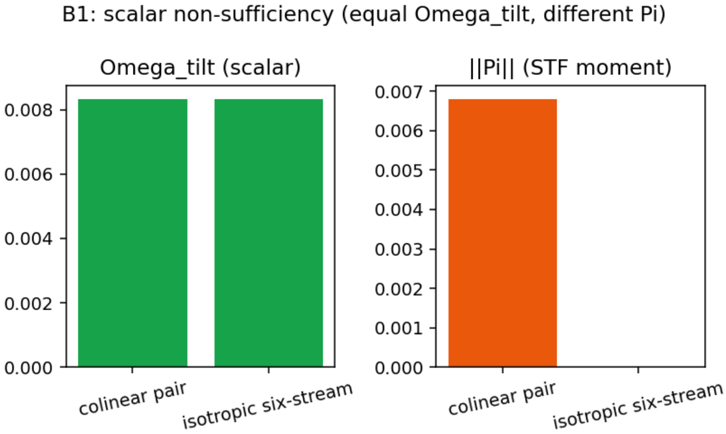

Figure 6: B-nonsuff: a colinear pair and an isotropic six-stream share Ω_{tilt} (left) but differ in $\|\Pi\|$ (right); the scalar trace is non-sufficient. Diagnostic-only.

reconstruction is least constrained; we report the reversal as a descriptor and do not attribute a physical cause. This is a model-independent kinematic descriptor of the reconstruction; it is not a cosmological-frame-violation or Bianchi claim, and the bootstrap dispersion under-represents the true direction uncertainty (correlated cells) rather than being a full covariance.

5.3 K5 — CF4 bulk flow from the full group release

The full Cosmicflows-4 release (Tully et al. 2023; VizieR J/ApJ/944/94), bound through an added download-pipeline stage, supplies 38,053 group peculiar velocities with distance moduli, errors, and supergalactic positions. We estimate the weighted-Gaussian maximum-likelihood (generalized-least-squares) bulk flow $B = A^{-1}b$, $A = \sum_i n_i n_i^\top/\sigma_i^2$, $b = \sum_i n_i V_i^{\text{pec}}/\sigma_i^2$, with per-group velocity errors from the distance-modulus uncertainty plus a fitted intrinsic dispersion $\sigma_\star$ (reduced $\chi^2 \sim 1$), and calibrate the amplitude with a forward mock (Table 3). The inverse-Fisher covariance is *conditional* on this error model (it omits cosmic variance and unmodelled selection); the hierarchical random-effect and release-matched-mock extensions are PR08-002/003.

Table 3: K5: CF4 group bulk flow in depth windows. $|B|$ is the weighted-GLS amplitude; the apex is in Galactic (l, b); the last column is the forward-mock 1σ coverage of the injected amplitude (a value near 0.68 indicates a calibrated error model).

$d < R$ [Mpc]	N	$\|B\|$ [km/s]	apex (l, b) [deg]	mock coverage
60	subset	320 ± 7	$(296, 20)$	0.68
100		351 ± 5	$(293, 22)$	0.72
150		347 ± 5	$(293, 21)$	0.65
200		346 ± 5	$(293, 21)$	0.67
300		345 ± 5	$(293, 21)$	0.67

Findings. The bulk flow is stable at ~ 345 km/s toward Galactic $(l, b) \approx (293°, 21°)$ across the depth windows, with a forward-mock coverage near the nominal 68% (the error model is calibrated). This K5 amplitude is *not* in tension with the larger K4 value (~ 590 km/s, §5.2): the two are different estimands — K4 is the volume-averaged velocity of the CF4 reconstruction grid (which carries the reconstruction's own dynamics over a window), whereas K5 is a GLS bulk flow from the group peculiar velocities themselves. The amplitude and direction are consistent with the WF/CR CF4 reconstruction line; we note the CF4 bulk-flow amplitude/scale is itself *contested*

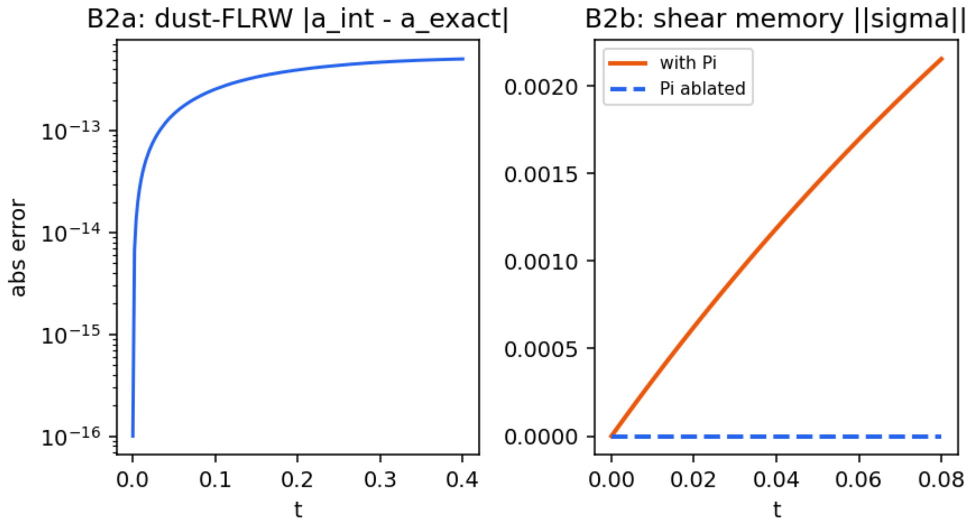

Figure 7: B-dust and B-shear: (left) the integrator reproduces the exact dust-FLRW scale factor to $\sim 10^{-13}$ over the test window; (right) the shear $\|\sigma\|$ grows with the physical anisotropic stress π_{ab} (equivalently the normalized $\Pi_{ab} = \kappa\pi_{ab}/(3H^2)$) and stays zero when it is ablated (shear memory). Diagnostic-only.

in the literature (a $\sim$4.8σ excess in some analyses versus footprint-systematic counter-analyses), so the comparison is to that reconstruction line, not a settled value. This is a diagnostic-only kinematic descriptor; $\sigma_\star$ is a fitted nuisance, the covariance is the inverse-Fisher matrix *conditional* on the error model (cosmic variance excluded), and selection/Malmquist biases are not fully forward-modelled. *No global-tilt or Bianchi-geometry claim is made from CF4 distance data alone.*

5.4 K6 — CF4 affine velocity-gradient decomposition

On the local CF4++ supergalactic velocity grid (128^3, 1000 Mpc box) we fit the affine flow $v = B + Mr$ in spheres about the observer and split the velocity-gradient tensor M into bulk B, expansion $\Theta = \mathrm{tr}\, M$, trace-free shear σ, and an antisymmetric (vorticity) part ω. The bulk amplitude rises from $\sim$152 to $\sim$422 km/s between 50 and 250 Mpc. Because the reconstruction is built from a curl-suppressing potential/Wiener field, the antisymmetric mode is removed *by construction*: physical vorticity is structurally non-identifiable from this field — it is not measured to be zero, and we therefore quote no recovered-ω number as an inference. The estimator's curl channel is nonetheless exercised by an algebra-only check: an injected solid-body rotation is recovered with relative error 3×10^{-16} (a mechanics test of the affine fit, not a vorticity measurement). The field-realization posterior and cross-reconstruction comparison remain *blocked* (`BLOCKED_MISSING_FIELD_REALIZATIONS`: only the CF4++ reconstruction is available).

Critically excluded candidates. Several nearby analyses were considered and *excluded* as empty or as re-presentations, to keep this report free of non-results: a joint common-axis posterior over the four dipoles (the available latent likelihood has no directional term, so the axis is prior-flat); the DESI footprint resultant-dipole "systematics floor" (already shipped as a long-run jackknife diagnostic); the DESI cosmological clustering dipole (blocked: no matched randoms/masks); and any positive Bianchi/tilt evidence (blocked: amplitude-matched contamination null, no native morphology atlas).

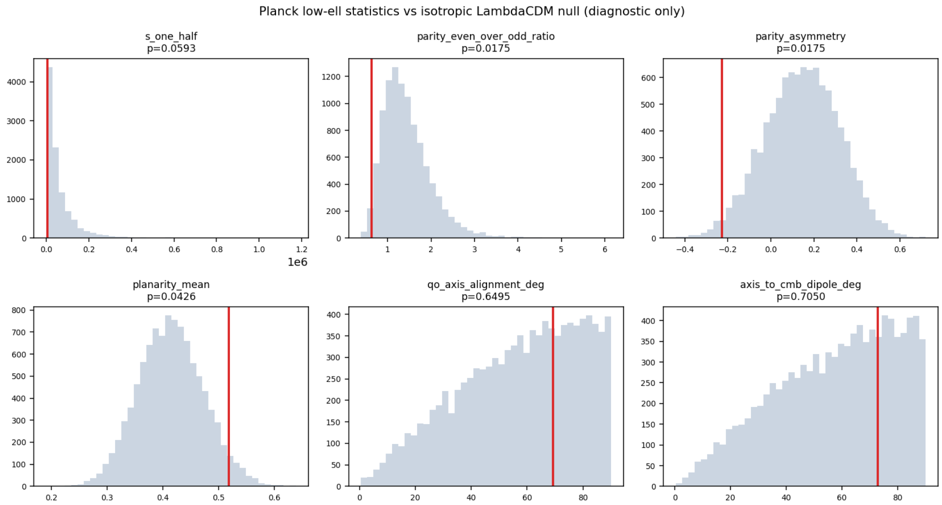

Figure 8: K1: observed low-ℓ statistics of the Planck PR3 map against the isotropic ΛCDM null ensemble. Model-independent null-calibrated features; not evidence for any Bianchi model.

6 Synthetic estimator-mechanics validations

The model-independent measurements above use estimators whose *mechanics* (coverage, bias control, look-elsewhere calibration, and structural identifiability) are validated separately on synthetic catalogues with known truth. These are explicitly *synthetic mechanics*, not measurements: they certify that the estimator behaves correctly, and they do not produce any observational number. The release-matched forward mocks, end-to-end simulations, and constrained field realizations needed to turn these mechanics into calibrated observational uncertainties are the standing blocks of §8.

K5 hierarchical bulk-flow likelihood (mechanics). A weighted-Gaussian hierarchical model $y = NB + W\alpha + Z\delta + \epsilon$, $\delta \sim \mathcal{N}(0, T)$, $\epsilon \sim \mathcal{N}(0, \mathrm{diag}(\sigma^2_{\mathrm{meas}} + \sigma^2_\star))$, profiles $\sigma_\star$ and absorbs method/group offsets as random effects. On 500 synthetic catalogues with a deliberately method-correlated angular selection (true bulk $(180, -95, 70)$ km/s, true $\sigma_\star = 145$ km/s), the naive diagonal fit undercovers the selection-aligned component (68% coverage 0.51) and carries a 15.0 km/s bias norm; the hierarchical fit restores component coverage to 0.67–0.69 (68%) and 0.94–0.96 (95%), reduces the bias norm to 1.9 km/s, and recovers $\sigma_\star = 142.7$ km/s (Table 4). This validates the random-effect mechanics that PR08-002 brings to the K5 estimator; the cosmic-variance and release-specific Malmquist terms remain blocked (PR08-003).

K6 curl-suppression no-go (mechanics). On 300 noisy synthetic affine fields carrying a known vorticity $\omega = (0.006, -0.004, 0.009)$, the full affine fit recovers it ($\bar{\omega} \approx (0.005, -0.005, 0.008)$), but projecting each field onto its symmetric (potential-flow) part before fitting erases the curl to $\leq 7.5 \times 10^{-17}$. A curl-suppressing reconstruction therefore makes physical vorticity *structurally non-identifiable* even when the pre-reconstruction field contains it—the formal basis for the K6 reporting discipline.

K1 max-scan global calibration (mechanics). For a frozen family of low-ℓ statistics evaluated in every simulation, each result is converted to a tail score, a single maximum is taken, and the global rank Monte-Carlo p-value (with the +1 correction) is formed. On 2000 synthetic correlated mocks the global $p = 0.0070$ is correctly no smaller than the most extreme local rank $p = 0.0015$, demonstrating the look-elsewhere mechanics that a real PR4/ NPIPE E2E summary (PR08-001, blocked) would require.

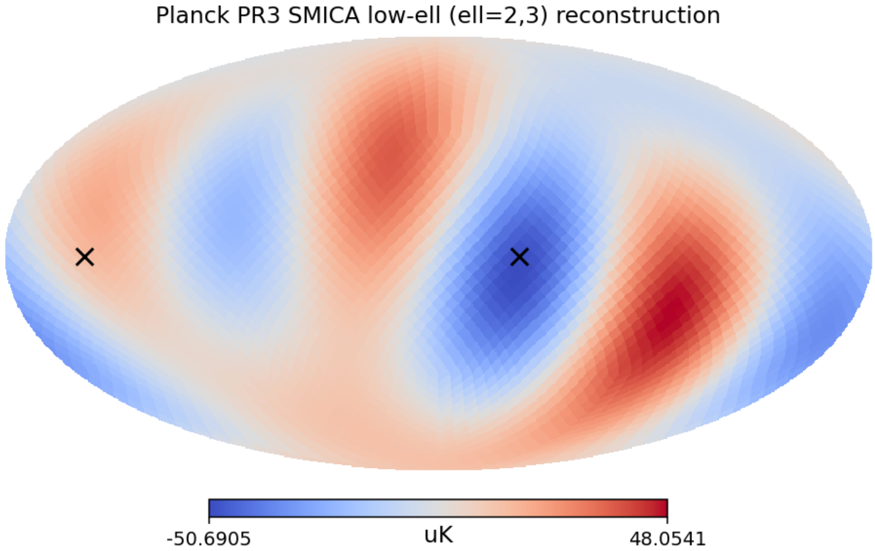

Figure 9: K1: the power-inertia low-ℓ ($\ell = 2, 3$) preferred axis of the real Planck map. The axis-alignment and axis-to-apex angles are null (Table 1). Diagnostic-only; a power-inertia-axis descriptor, not the multipole-vector axis-of-evil statistic.

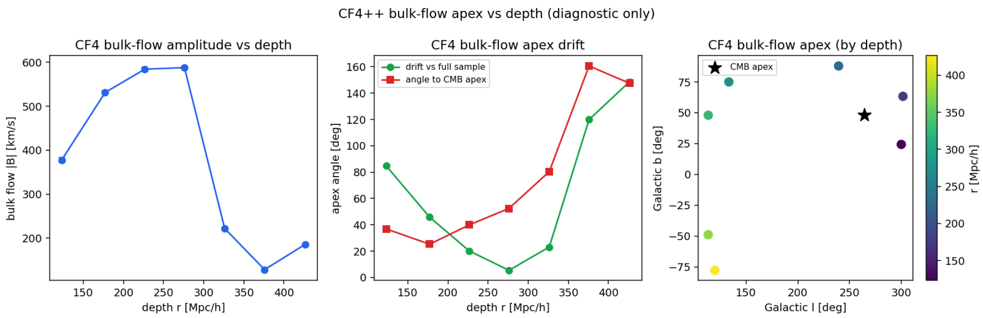

Figure 10: K4: CF4++ bulk-flow amplitude and apex drift versus depth shell. Model-independent kinematic descriptor; not a frame-violation or Bianchi claim.

7 Transfer-conditional diagnostics

A separate family of result surfaces depends on *external* or *empirical-proxy* transfer functions (AniCLASS-external and an empirical shear$\to D_\ell$ proxy). These are *transfer-conditional* diagnostics: they are not native low-ℓ solver output, they carry no passed native-validation gate, and the family labels attached to them are provenance labels only, not classification claims. Any amplitude-ranking quantity computed along these paths (for example a configured likelihood-ratio summary for an admitted dipole-amplitude premise) is a premise-conditioned amplitude fit and a transfer-conditional ranking; it is prior- and error-budget-sensitive, it is not a source-identification claim, and the source origin is not established. The auditable inventory of transfer dependence is `docs/generated/transfer_sensitivity_report.md`; the calibration window for the external shear$\to D_2$ path is the legacy Saadeh-linear regime $\Sigma^2 \in [10^{-24}, 10^{-20}]$. These rows are included for completeness and provenance, not as headline results.

8 What is not claimed (standing blocks)

For the avoidance of doubt, the following are *not* results of this report and remain blocked:

- an identified Bianchi family or a detected anisotropic geometry from any scalar, axis, or low-ℓ

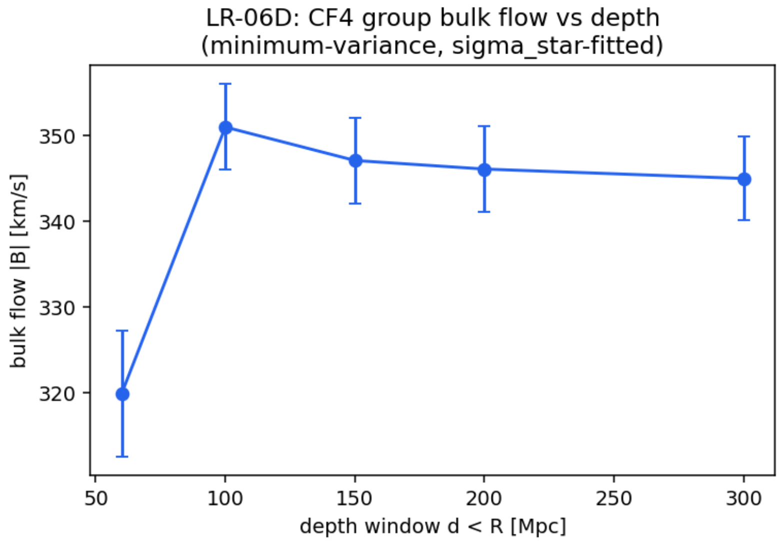

Figure 11: K5: CF4 group bulk-flow amplitude versus depth window (weighted-GLS, $\sigma_\star$-profiled, conditional 1σ error bars). Diagnostic-only kinematic descriptor; not a frame-violation or Bianchi claim.

Table 4: Synthetic estimator-mechanics summary (known truth; no observational number). All rows are *synthetic mechanics*.

Mechanism	Quantity	Naive / pre	Repaired / post
K5 hierarchical	worst 68% component coverage	0.51	0.67
K5 hierarchical	bias norm [km/s]	15.0	1.9
K5 hierarchical	recovered $\sigma_\star$ [km/s]	—	142.7 (true 145)
K6 curl no-go	projected $\|\omega\|$	$\sim 10^{-2}$ (full)	7.5×10^{-17}
K1 max-scan	global vs. min-local p	—	$0.0070 \geq 0.0015$
B-dynamics	1000-state E/p residual	—	$< 6 \times 10^{-14}$

feature above;

- native-solver validation for any external/proxy transfer output;
- any MIO observatory certificate used as a posterior, a truth certificate, a model weight, or an HTT evidence term;
- any low-ℓ or velocity feature above promoted from a model-independent measurement into anisotropy evidence;
- a globally significant low-ℓ anomaly detection (the K1 p-values are local, look-elsewhere-tracked, single-sky, and not full-covariance/mask-coupled);
- any calibrated observational uncertainty from the synthetic estimator-mechanics of §6: the K1 global p-value needs PR4/NPIPE end-to-end summaries (`BLOCKED_MISSING_PR4_E2E_ACCESS`); the total K5 uncertainty needs release-matched forward mocks and cosmic variance (`BLOCKED_MISSING_RELEASE_MOCK_OWNERSHIP`); the K6 vorticity/shear posterior needs constrained field realizations (`BLOCKED_MISSING_FIELD_REALIZATIONS`);
- any family-conditioned CMB likelihood, transfer-based compatibility/exclusion, or global-tilt posterior from CMB transfer: these await the separate native low-ℓ Bianchi-Boltzmann solver (exact-FLRW comparator first), and old external/Rust spectra are not admissible substitutes (`AWAITING_NATIVE_LOWELL_SOLVER`).

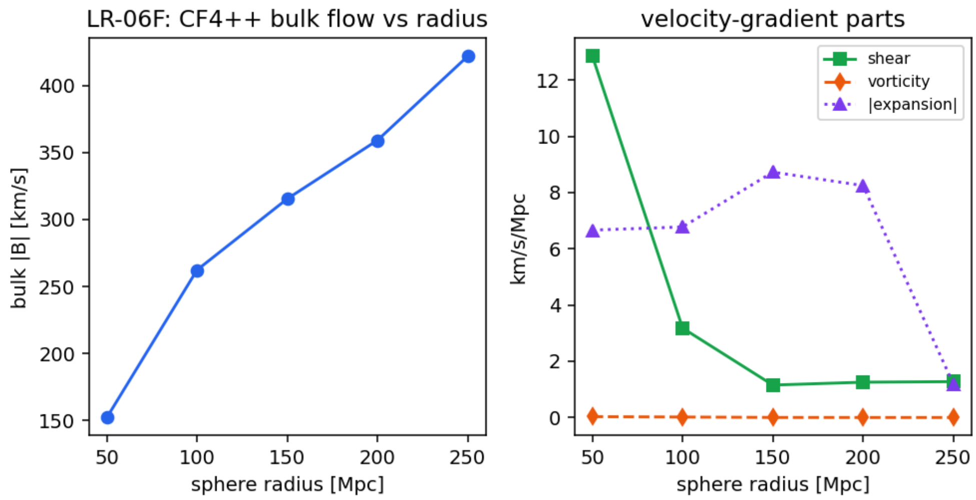

Figure 12: K6: (left) CF4++ bulk flow versus sphere radius; (right) the velocity-gradient parts—shear, vorticity (reconstruction-suppressed), and $|\Theta|$. Diagnostic-only; vorticity is reconstruction-conditioned.

9 Reproducibility

Every result above is regenerated by a recorded command and ships a manifest with input hashes and a claim tier. The load-bearing artifacts are:

- **EGS-type theorems:** `python scripts/prove_egs_lowell_theorems.py` $\to$ `docs/generated/egs_lowell`
  figures via `python scripts/make_egs_lowell_theorem_figures.py`.
- **PAPER-A/B theorems:** `make paper-a-gates` / `make paper-b-gates` ($\to$ `docs/generated/pr04_paper_`
  figures `python scripts/make_pr04_paper_figures.py`).
- **K1 (Planck):** `python scripts/make_lowell_morphology_real_map.py` $\to$ `docs/generated/lowell_morph`
- **K4 (CF4 apex):** `python scripts/make_cf4_bulkflow_apex_depth.py`.
- **K5 (CF4 full release):** `python dl_pipeline/scripts/fetch.py --stages cf4_full` then `python scripts/make_cf4_bulkflow_likelihood.py`.
- **K6 (CF4 affine flow):** `python scripts/make_cf4_affine_flow.py`.
- **Transfer inventory:** `python scripts/generate_transfer_sensitivity_report.py`.
- **PR07 repair gates + symbolic/CoVe surface:** `make pr07-gates` (unit-safe stress, independent dynamics, PAPER-A closure); `make pr07-wolfram` $\to$`docs/generated/pr07_wolfram_proofs.json`
  (local Wolfram/xAct); `python scripts/run_pr07_experiments.py` then `python scripts/cove_verify_pr0`
  $\to$`docs/generated/pr07_{paper_a,paper_b,k1,k5,k6}.json` and `pr07_cove_report.json`.
  The step-by-step PR/blocker registry is `docs/research_program/pr07/`.

Each figure in this report has a sidecar `*.manifest.json` recording owner, claim tier (*diagnostic-only*), `family_identification:false`, and `native_solver_result:false`.

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